Quick Start Guide
FLIM Data Acquisition Card

FL-DAQ-02
FL-DAQ-04
FL-DAQ-08

1 About This Guide

This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at: https://www.flimlabs.com/products/FLIM-DAQ/

Scan the QR code to open the full manual.

Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use

The FLIM Data Acquisition Card is intended as a compact electronic module for acquisition, processing, and transmission of electronic signals by professional users in private research laboratories, industrial environments, or universities. It may be used as a stand-alone unit or integrated into custom measurement systems and scientific instrumentation.

The device must be operated only by qualified personnel familiar with laboratory electronic instrumentation.

4 Product Overview

The FLIM Data Acquisition Card is a compact FPGA-based electronic module designed for the acquisition, processing, and transmission of high-frequency digital signals. It receives digital input signals from external devices, processes them deterministically, and transfers the acquired data to a host computer via USB for analysis.

The device is optimized for high-speed timing applications such as spectroscopy, fluorescence lifetime measurements, and time-correlated photon counting systems.

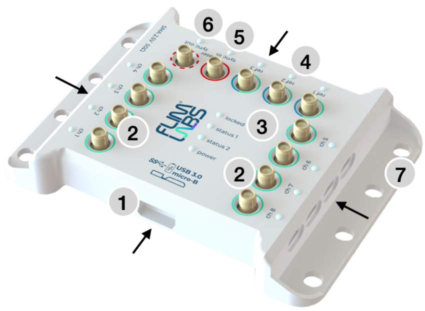
Key Features

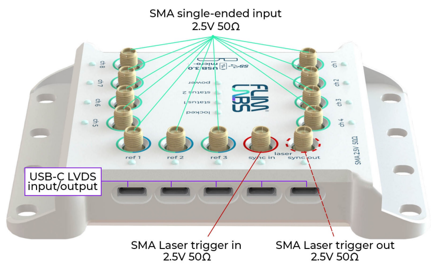
- FPGA-based digital signal processing
- SMA LVTTL and USB Type-C LVDS input interfaces
- Dedicated laser trigger input and output for synchronization
- USB interface for power, configuration, and data transfer
- Compact laboratory form factor
- Variant-dependent number of input channels (DAQ-02 / DAQ-04 / DAQ-08)

5 Components Overview

Overview of the key components of the device.

1. USB micro-B port for power supply and data transmission to the host computer
2. Signal input ports via SMA connectors and USB Type-C interfaces
3. Status LED indicators for power and operational states
4. Reference signal ports via SMA connectors and USB Type-C interfaces
5. Laser synchronization input port
6. Laser synchronization output port
7. Mounting holes on the enclosure





6 Scope of Delivery

- Data acquisition card (DAQ)
- USB 3.0 SuperSpeed micro-B cable
- Quick Start Guide
- Declaration of conformity
- Software for acquisition and analysis available from the manufacturer website

7 Setup

Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- verify that the input signal meets the requirements given in Table 1

8 Connections

USB connection (power and data)

- Connect the device to the host computer using the USB 3.0 micro-B cable.
- The USB connection provides both 5 V DC power supply and data transfer to the host PC.
- Verify that the status LEDs indicate normal device status.

Input signal connections

- Connect external LVTTL signals to the SMA inputs using compatible 50 Ω coaxial cables.
- Alternatively, connect LVDS signals using compatible USB Type-C signal cables.

Laser trigger and synchronization connections

- Connect the external laser trigger signal to the laser trigger input port when required by the application.
- Connect the laser trigger output port to external devices requiring synchronization.

9 Configuration

The device can be operated using the software applications *Spectroscopy Py* or *FLIM Imager*. If the device is not detected automatically after starting the software, use the *Check device* function to initiate a manual device search.

The software allows automatic channel identification using the *Detect channels* function or manual selection of the desired input channels. For each selected channel, verify that the connection type corresponds to the physical interface used (USB Type-C or SMA).

10 Operation

Once the power supply, input signal, and synchronization connections have been correctly established and the initial configuration has been completed, the device operates under control of the host software. The software configures the acquisition parameters, controls the start and stop of measurements, and displays the acquired data in real time. Please refer to the respective software manuals for further information. An example measurement setup is shown in Figure 1.

11 Further Information

Detailed information on troubleshooting, maintenance, storage, and disposal is provided in the full operation manual available at https://www.flimlabs.com/products/FLIM-DAQ/.

12 Product Label

FLIM Data Acquisition Card

P/N: FL-DAQ-08
S/N: FL-DAQ-XXXX-XXX
5 V ⎓ 0.9 A | IP40

MADE IN ITALY - **FLIM LABS S.r.l.**
Via della Farnesina 3 - 00135 Rome
techsupport@flimlabs.com

13 Technical Specifications

Item	DAQ-08	DAQ-04	DAQ-02
Electrical Specifications			
Electrical safety class	Class III		
Power supply	5 V DC via USB micro-B		
Maximum current consumption	0.9 A		
Maximum power consumption	4.5 W		
Signal Specifications			
Input polarity	positive		
Minimum input pulse width	> 1.5 ns		
Minimum input pulse height (abs)	2.5 V		
Maximum input pulse height (abs)	5 V		
Single-ended SMA inputs (LVTTL)	11	7	5
USB Type-C LVDS inputs	12	8	6
Impedance	50 Ω		
Minimum resolvable fluorescence lifetime	50 ps		
Single-event precision ($\sigma/\sqrt{2}$)	300 ps		
Supported laser sync frequencies	Input: 10 MHz to 80 MHz Output: 10, 20, 40, 80 MHz		
Maximum transfer rate (sum over channels)	100 M counts/s		
Maximum burst rate per input channel	640 M counts/s		
Physical Specifications			
Dimensions (L x W x H)	139 mm x 101 mm x 28 mm		
Weight	120 g		
Protection rating	IP40		
Environmental Specifications			
Operating temperature	0°C to +40°C		
Storage temperature	-20°C to +70°C		
Relative humidity	Up to 80% non-condensing		

Table 1: Technical specifications of the FLIM Data Acquisition Card.

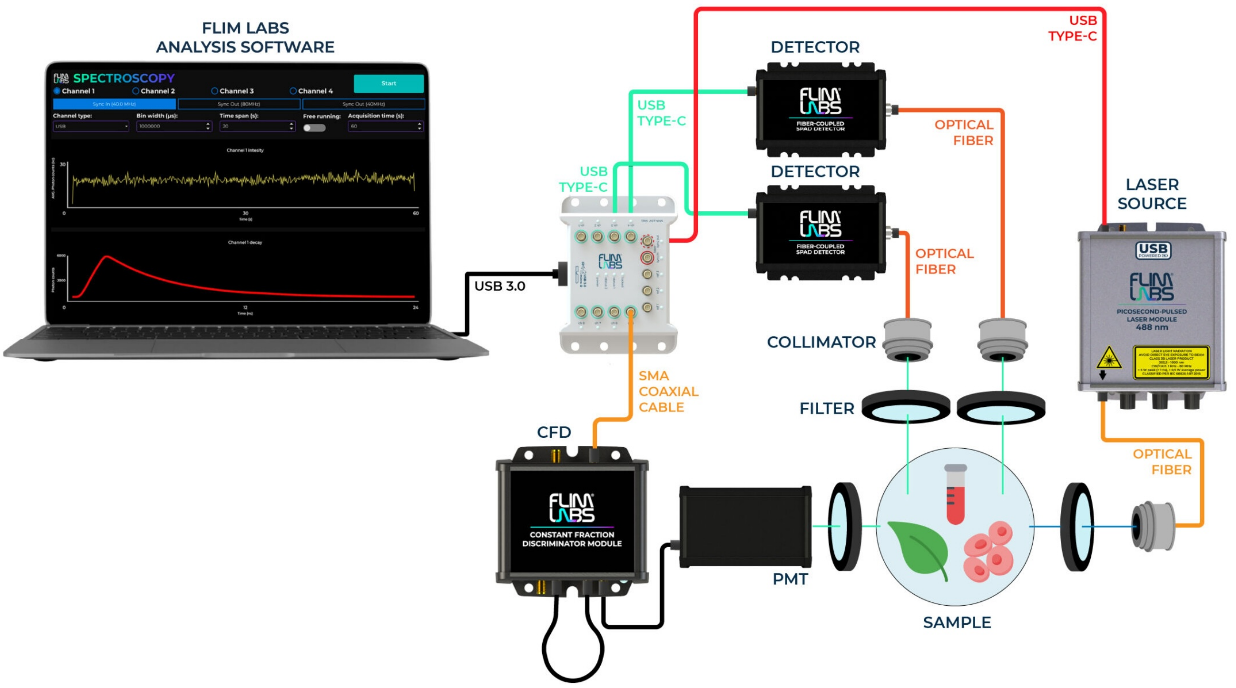


Figure 1: Example configuration of a fluorescence lifetime analysis system using multiple detectors.